Family Violence in Victoria

Data Visualisation Report

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1 Problem Description and Motivation

For this project, I will be investigating family violence (FV) in Victoria. Understanding where family violence is most prevalent can help support organisations allocate funding where it is most needed. Furthermore, understanding how family violence crime differs from non-FV crime, and how rates of family violence have changed over time can help to identify initiatives that may reduce incidence of family violence.

The research questions I aim to answer are:

1. How has family violence in Victoria changed over time?
2. Which LGAs have the highest rates of each family violence crime?
3. How do the rates of family violence related crimes compare to those of similar non-FV crimes?

The data I used to answer the above questions came from the Victoria Crime Statistics Agency website. I used two datasets, one containing yearly crime rates by LGA and offence type from 2016 to 2025, and a spatial dataset containing polygons for LGAs in Victoria. These datasets can be found at the following links:

`https://files.crimestatistics.vic.gov.au/2025-12/Data_Tables_LGA_Recorded_Offences_Year_Ending_September_2025.xlsx`

`https://datashare.maps.vic.gov.au/search?q=uuid%3Dcff349f9-3ef3-51bd-b5a9-ae650f6d6c2`

2 Data Checking and Wrangling

The dataset I used for this project contained ten attributes. These were year, year ending (ie, the month that was taken as the end of the year), police service area (PSA), local government area (LGA), offence division, offence subdivision, offence subgroup, offence count, PSA rate per 100,000 population, and LGA rate per 100,000 population.

To clean this dataset, I first removed columns that I didn't need. I wanted to look at offence rates by LGA, so I didn't need the police service area, offence count, or PSA rate per 100,000 population columns. I also removed the offence division and offence subdivision columns, as the offence subgroup column contained the most detailed offence types. For simplicity, I renamed the local government area column to 'LGA', the LGA rate per 100,000 population column to 'Rate', and the offence subgroup column to 'Offence'.

Next, I cleaned the offence column. The values in this column contained an offence code, followed by the offence name. I first removed the code, then selected only rows where the name contained either 'FV' or 'non-FV'.

I then created a new column called 'Type' which contained either the value 'FV' or 'non-FV', based on the offence column. I also removed the prefix of 'FV' or 'non-FV' from all values in the offence column.

Next, I changed the name 'Colac-Otway' in the LGA column to 'Colac Otway', to match the formatting in the spatial dataset. I also changed all values in the LGA column to uppercase, again to match the spatial dataset.

Lastly, I noticed that the dataset didn't have a row for every possible combination of Year, LGA, Offence, and Type. This is a problem, because I wanted to be able to make a choropleth map for each year and offence. As I was unable to source the missing data, I instead added a row with Rate set to 'NA' for each missing combination of the other attributes. This meant that when I created a choropleth map, I could set the NA values to a specific colour, rather than LGAs with no data available simply disappearing from the map altogether.

3 Data Exploration

To aid in the analysis of my data, I calculated a number of derived variables. Firstly, I calculated offence rates for the whole state, by adding rates from all LGAs with the same Year, Offence, and Type values. I also calculated the total rate of FV offences and total rate of non-FV offences, for each LGA and for the whole state, using a similar method (Bobbitt 2021).

Next, I calculated the increase in offence rates from 2016 to 2025. To do this, I simply extracted the 2016 and 2025 data from the full dataset, and subtracted the 2016 rates from the 2025 rates. I did this both for each LGA, and for the whole state.

The first exploratory visualisation I made was a series of choropleth maps for each year, displaying the total FV offence rate in each LGA. This visualisation can be seen in figure 1, and clearly shows that the total FV offence rate changes over time for most LGAs. There also doesn't seem to be any clusters of LGAs that consistently have very high or very low offence rates. However, it seems as though the LGAs with very high offence rates tend to be in rural areas, not close to Melbourne.

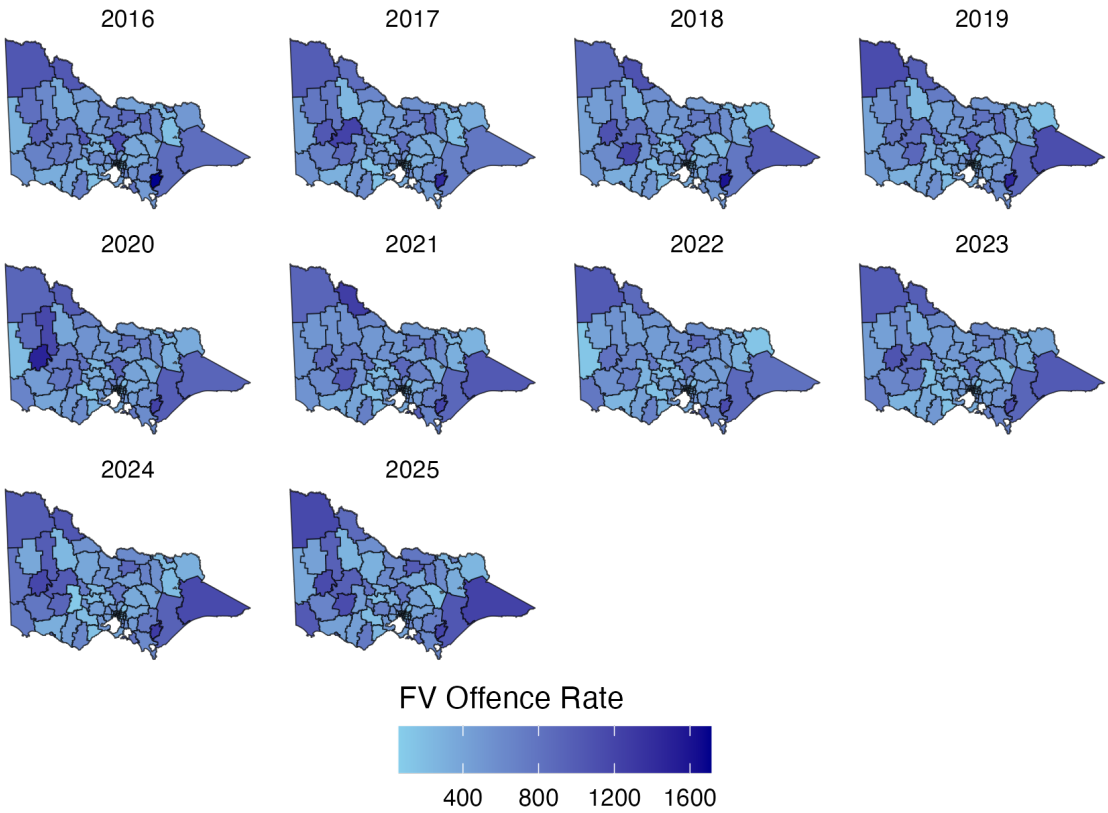


Figure 1: Map of total FV offence rate by year and LGA

I also made a series of choropleth maps showing offence rates by type and offence for the year 2025, shown in figure 2. In this figure, the grey coloured LGAs correspond to those for which offence rate data was unavailable. Since this figure uses the same colour scale for each plot, it is easy to see that the rates of FV common assault are clearly the highest of all the offence types. The offence rates for harassment and private nuisance and stalking are both quite low, and roughly constant across different LGAs. However, it does seem that the LGAs with higher rates of these offences are those further from Melbourne, covering rural areas. Comparing the FV and non-FV maps for common assault and serious assault, two things are quite evident. Firstly, the FV rates are generally higher than the non-FV rates for the same offence. Secondly, the LGAs with high FV rates are generally in rural areas, leading to the conclusion that family violence is perhaps more prevalent in rural areas.

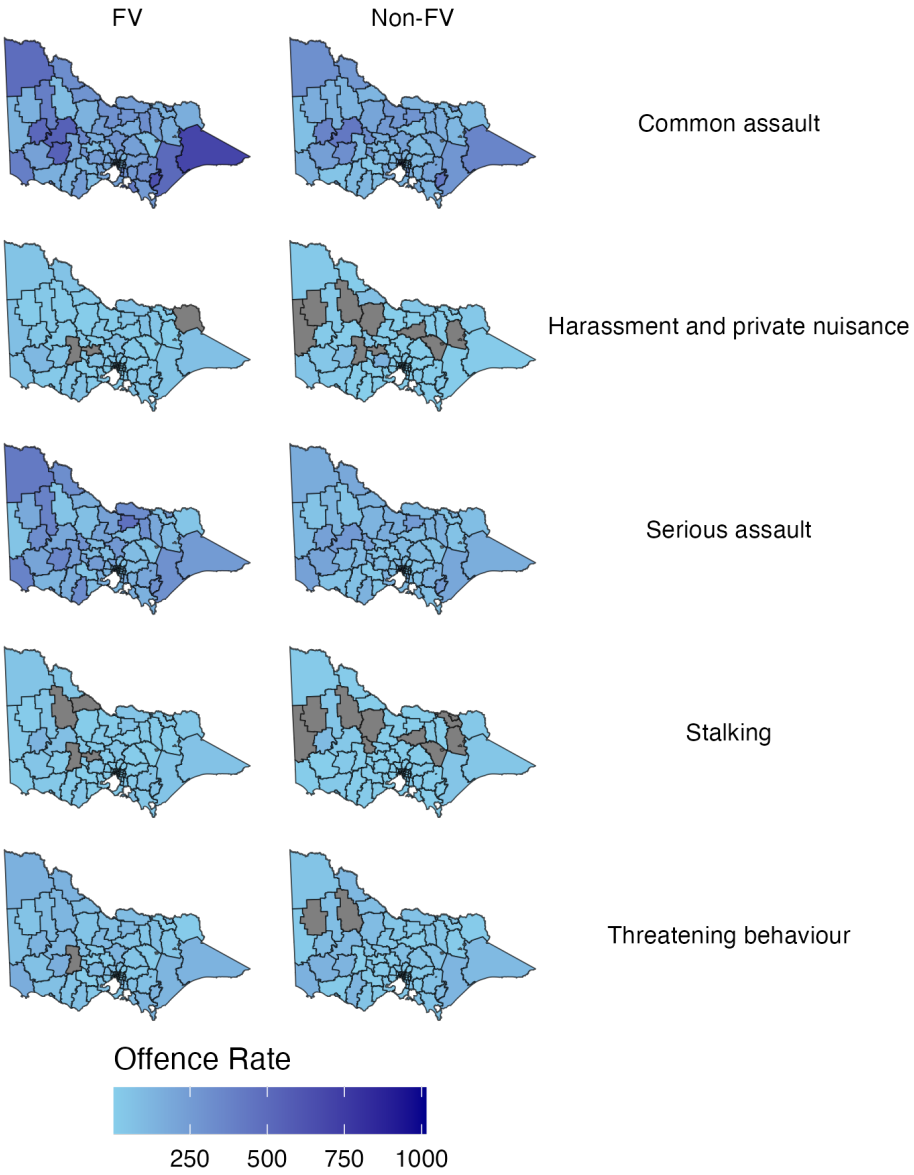


Figure 2: Map of offence rates by type and offence for 2025

Figure 3 shows the last choropleth map I made, which displays the increase in total FV offence rate from 2016 to 2025. It seems that in general, LGAs that are closer to Melbourne have had less change in the FV offence rate between 2016 and 2025.

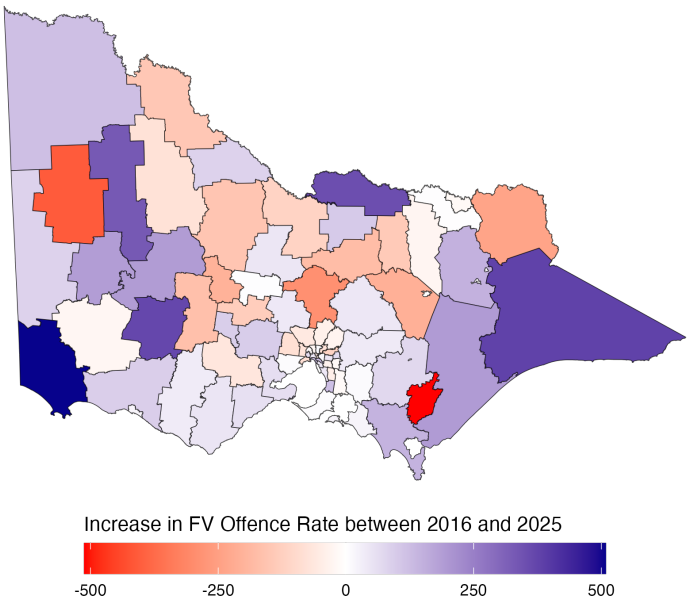


Figure 3: Map of increase in FV offence rate between 2016 and 2025, by LGA

The next plot I made is a bar chart showing the increase in offence rates for difference offences between 2016 and 2025, for the whole state of Victoria. This plot is shown in figure 4. The most obvious feature of this graph is that non-FV common assault has increased significantly, while non-FV serious assault has decreased significantly. Without knowing much about criminal law in Victoria over the past decade, I could draw the conclusion that perhaps the definitions of common assault and serious assault were modified at some point, resulting in some offences that were previously categorised as serious assault now being considered common assault. However, I am not interested in the raw increase in non-FV crimes, only how they related to the increase in FV crimes.

Looking only at the FV crime rates, they have all increased in the past decade, except for threatening behaviour. One point of concern is that the rates of FV serious assault and FV stalking have increased, while their non-FV counterparts have decreased. This suggests that initiatives that effectively reduce non-FV crime may not work for FV crime, and different solutions should be used.

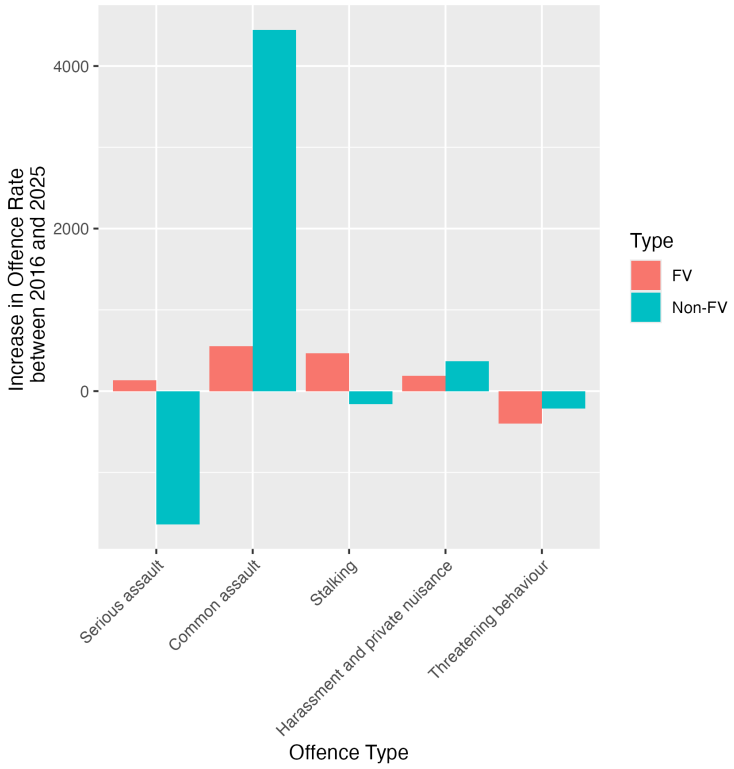


Figure 4: Bar chart showing increase in offence rates between 2016 and 2025, for the whole state of Victoria

The next plot I made is a pair of scatterplots showing offence rates for each of the different FV and non-FV crimes over time for the whole state of Victoria, in figure 5. The first thing that is clear from this plot is that the rates of each offence are similar for the FV and non-FV categories of the same offence, except for common assault. The rates of FV common assault are much higher than the rates of non-FV common assault.

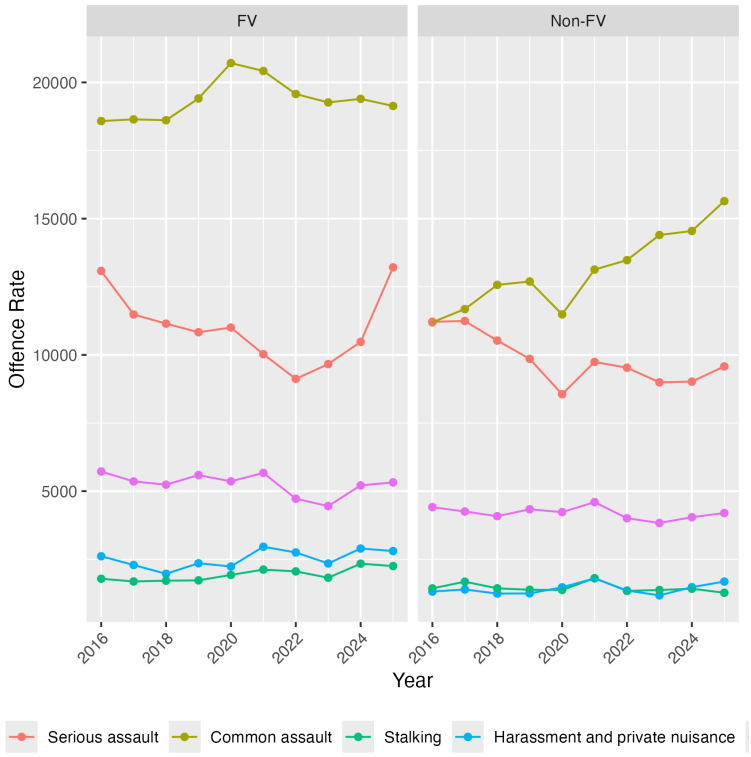


Figure 5: Scatterplot showing offence rates over time for both FV and non-FV offences, for the whole state of Victoria

I also plotted the total offence rates for FV and non-FV over time for each LGA, shown in figure 6. While the individual plots are quite small, making it difficult to extract detailed information, there are some conclusions that can be drawn. Firstly, most LGAs have similar rates of FV crime and non-FV crime. The one LGA where the rate of FV crime differs the most from the rate of non-FV crime is Melbourne.

It can also be seen that for most LGAs, the rates of FV and non-FV crime stay roughly constant over the ten year period. One LGA where this is clearly not the case is Latrobe. The FV crime rate in Latrobe has steadily decreased since 2016.

Another interesting feature is that a few LGAs have a sudden spike in FV crime, where non-FV crime remains roughly constant. For example, Horsham and Yarriambiack have this spike in 2020, and Northern Grampians has it in 2017. The spike in 2020 can likely be attributed to the covid pandemic. Lockdowns resulted in people spending more time with their families, so those already committing family violence might commit these offences more often. It would be interesting to see what caused the spike in the Northern Grampians in 2017.

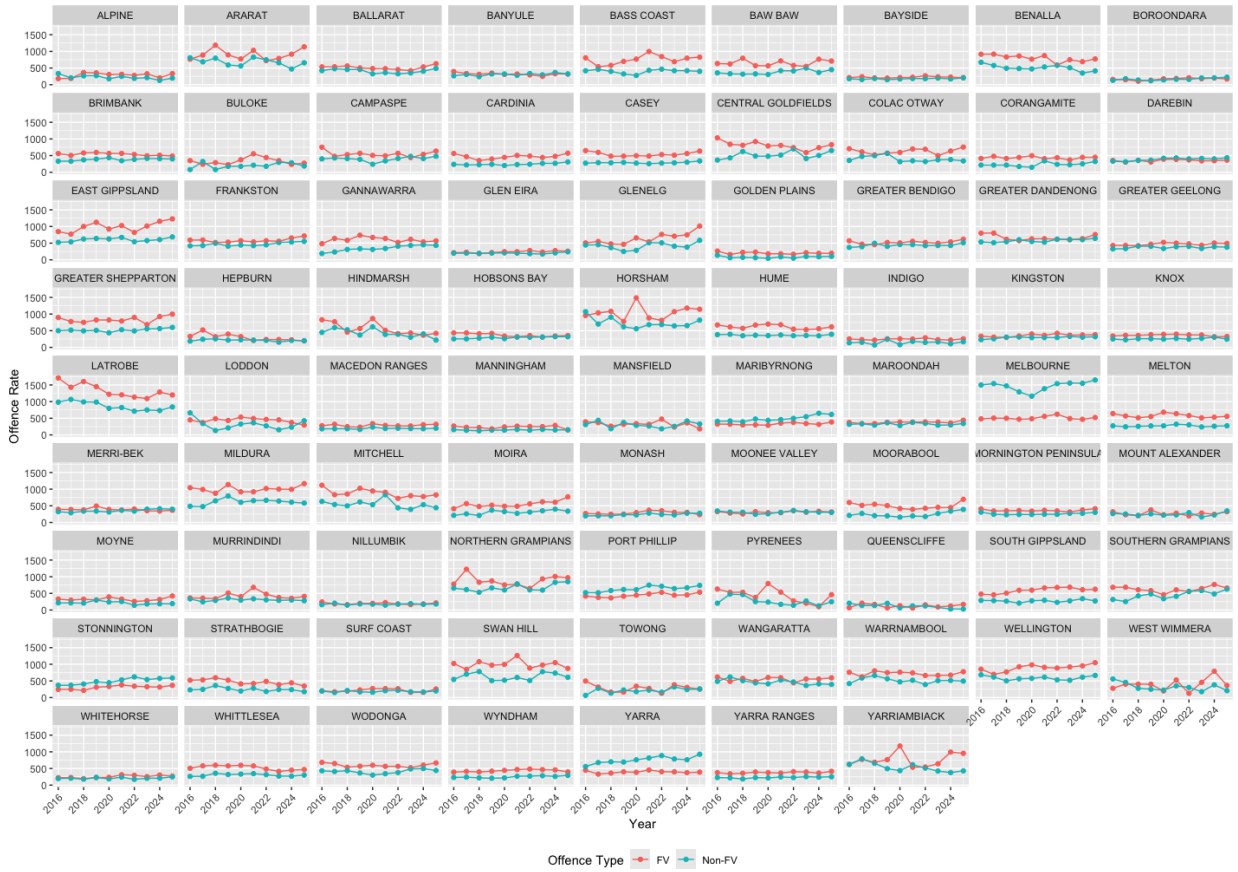


Figure 6: Scatterplots showing total FV and non-FV offence rates over time, by LGA

The main insights I have gained through my exploratory data analysis are 1. FV common assault is much more prevalent than non-FV common assault, whereas other offences have similar rates for the FV and non-FV categories 2. FV is more prevalent in rural areas 3. Most types of FV crime have increased over the past decade

In designing my narrative visualisation, I wanted the user to be able to discover these insights on their own through interacting with the visualisation. Therefore, I chose to include a map showing the geographic spread of offence rates in LGAs, as well as a scatterplot showing how offence rates have changed over time. I added interaction to allow users to choose which offence to show on both the map and scatterplot, so they can compare the plots for different offences. I also allowed users to select the year to show on the map so they can compare the geographic distribution across different years. Finally, I added the option to have the map display the increase in offence rates from 2016 to 2025, rather than the offence rate for a given year. This allows the user to clearly see where FV has changed the most and least.

4 Exploration Findings, Audience and Data Story

There were three key findings from my data exploration. FV common assault is far more prevalent than non-FV common assault, whereas other the rates of other offences don't differ significantly between the FV and non-FV categories. FV is generally more prevalent in rural LGAs. Most types of FV crime in Victoria have increased over the past decade.

The main message I would like my visualisation to convey is that FV generally increasing, and that the LGAs most in need of intervention are those in rural areas.

My intended audience is policy makers and family violence support organisations. These are the people with the ability to provide support services to victims of family violence and those with the power to implement initiatives to reduce family violence.

These people should be familiar with interpreting data, so a data visualisation that allows the user to extract insights themself though interactions is appropriate.

5 Visualisation Design Process

During the initial brainstorming process, most of my ideas fell into one of three categories: map, scatterplot, and bar chart. My first idea was a choropleth map displaying the total FV offence rate for each LGA, with a pie chart overlayed on each LGA showing the proportion of each different FV offence type for that LGA. I also considered three other map visualisations. The first was a set of two side by side choropleth maps, one displaying FV offence rates and the other displaying non-FV offence rates, allowing for comparison between the two. Next was a choropleth map displaying offence rate, with an interaction menu allowing the user to select offence type, year and FV or non-FV. I also considered a choropleth map displaying the increase in offence rate between 2016 and 2025, with the user able to select offence type.

I considered two different bar charts. The first had offence type on the x-axis, and offence rate on the y-axis, using colour to distinguish between FV and non-FV offences. The second bar chart I thought of was similar, but instead of showing offence rate, it shows the increase in offence rate between 2016 and 2025, again with offence type on the x-axis and colour distinguishing between FV and non-FV offences.

I only considered one scatterplot, showing offence rate over time for the whole state, with a different line for each offence type. A different colour would be used for each offence, and point shape would be used to distinguish FV from non-FV.

The final idea I had was a pie chart, showing the proportion of different offences out of the total rate of FV crime.

For my first full design, shown on sheet 2 in the appendix, I decided to combine the choropleth map showing the total FV offence rate with pie charts overlayed over each LGA, and a scatterplot showing how the total offence rates for the whole state changed over time. My original idea for the map had the pie charts constantly present, however I thought this might look a bit cluttered and therefore be difficult to interpret. To simplify the visualisation, I instead decided to have a pie chart pop up over an LGA when a user clicks on or hovers over that LGA. This allows users to view the whole map without the pie charts, and view proportions of different offence types in LGAs of interest by interacting with the visualisation. I also decided to include a sliderbar to allow the user to select which year to display on

the map. One clear downside to this visualisation is that there is no interaction with the scatterplot. Furthermore, the scatterplot only shows offence rates for the state as a whole, not for individual LGAs. Therefore, a potential refinement is to have the scatterplot display offence rates for a single LGA when a user clicks on an LGA.

My next alternative design, shown on sheet 3 in the appendix, combines an interactive choropleth map showing offence rates, and a bar chart showing the increase in offence rate between 2016 and 2025 on the y-axis, and offence type on the x-axis. The bar chart also uses colour to distinguish between FV and non-FV offences. The interaction available for the map includes allowing the user to select the offence type to display, and the year to display. The user can also zoom in on the map and click or hover over an LGA to reveal a pop-up box displaying the LGA name and offence rate for the data currently shown. Again, a clear downside is that there is no interaction with the bar chart, and no clear visual connectivity between the two charts.

My last alternative design, shown on sheet 4 in the appendix, involves a choropleth map displaying the increase in offence rate between 2016 and 2025, and a bar chart with offence rate on the y-axis and offence type on the x-axis. Again, the bar chart uses colour to distinguish between FV and non-FV offences. The user can select the offence type to be shown on the map, and the year to be shown on the bar chart. One downside of this visualisation is that the bar chart only shows the offence rates for the whole of Victoria, not each individual LGA. Again, this could be solved by having the bar chart display data for an individual LGA when a user clicks on an LGA.

For my final design, shown on sheet 5 in the appendix, I combined the maps from sheets 3 and 4, along with a simplified version of the scatterplot from sheet 2. To combine the maps, I added interaction to allow the user to choose whether they want to show the offence rate or the increase in offence rate from 2016 to 2025 on the map. I did this because I thought that both maps provided valuable insights, and together the user can gain a deeper understanding of the data. I also kept the interaction enabling the user to select the offence type and year, although the year selection is only applicable when offence rate is selected as the map type.

The main visual variable I used on the map was colour. For the offence rate choropleth map, the rate encoded by colour will always be positive, therefore I used a gradient between two colours, sky blue and dark blue. For the increase in offence rate map, the increase encoded by colour can either be positive or negative. Therefore, I used a gradient between three colours, red, white and dark blue. This resulted in the negative values ranging from peach to red, and positive values ranging from light purple to dark blue, with a value of zero represented by the colour white. It is good that the positive values showed up on the map as purple rather than blue, as it ensures users understand that this map shows different data than the other choropleth map. I chose red and purple, as they are very easy to distinguish, so it is easy to tell which values are positive and which are negative by even just a quick glance at the map.

I chose to include a scatterplot, as I believe it is very beneficial to be able to see offence rates for all years on the one chart. However, the scatterplot I used in sheet 2 was a bit cluttered and difficult to interpret. Therefore, I decided to simplify it to only show the offence rate for one offence. This meant the plot only had two lines, one for the FV offence, and one for the non-FV offence. These two lines are distinguished by colour, as it is an easier

visual variable to interpret than point shape or line type. The two colours I chose are red and blue, which are very clearly different and therefore easy to distinguish.

The default state of the scatterplot is to show the offence rate for the whole state. However, when a user clicks on an LGA, the scatterplot will display data for that LGA only. I also included a button to reset the scatterplot to show the whole state data.

6 Implementation and Technical Decisions

To implement my visualisation, I used RShiny. I used a sidebar to hold the interaction menus. On this sidebar, I first inserted a drop down menu allowing the user to choose the map type, either offence rate or increase in offence rate from 2016 to 2025. Next, I included a drop down menu allowing the user to select offence type, which is applied to both the map and scatterplot. Below this was a sliderbar allowing the user to select a year, which is only applicable when 'offence rate' is selected as the map type. Finally, I included a button allowing the user to reset the scatterplot to show data for the whole state of Victoria.

To create the map, I used leaflet. This allows the user to easily zoom in on the map and move the map around to centre it on an LGA of interest to the user. I chose to make the LGA fill colours slightly transparent so that the user can still see city names, roads and other features of the underlying map. Using leaflet also allowed me to implement the other interactions mentioned in the previous section (Cheng et al. 2023b). When a user hovers over an LGA, the LGA name and offence rate (or increase, depending on the map type selected) are displayed in a pop-up box beside the cursor. When they click on an LGA, the scatterplot shows data for this LGA only. To create the scatterplot, I used the ggplot2 package.

The last feature I added is a line of text above the map clearly and concisely explaining what data is being shown, and a similar line of text above the scatterplot. This ensures that users don't get confused about what data is currently being displayed.

The main difference between my final design in the appendix and my implemented product is the position of the interaction menus. Originally, I wanted to have the interaction menus along the bottom of the visualisation. However, RShiny makes it very easy to implement these features along the left side in the form of a side bar. I also think this visually makes it easier for a user to understand, as there is a vertical line separating the menus from the charts themselves (Shiny n.d.).

7 User Instructions

In order to run the code for my visualisation, the following packages are required: tidyverse, dplyr, readr, sf, ggplot2, leaflet, scales, stringr, shiny, and bslib.

My code also assumes that the working directory is the directory containing the R file. Furthermore, it assumes that the data required is in a folder called 'Project Data' inside the working directory.

Interacting with the visualisation does not require much explanation, as the visualisation itself is mostly self-explanatory. You can use the option menus on the sidebar to select data,

zoom in on the map, and click on LGAs to filter the data shown on the scatterplot.

8 Conclusion and Reflection

In conclusion, the data analysis performed for this project has allowed me to draw the conclusions that family violence is generally more prevalent in rural LGAs than those closer to Melbourne. Furthermore, common assault is the most prevalent family violence crime, and most types of family violence crime in Victoria have increased over the past decade. My visualisation allows a user to explore the data through interaction and draw these conclusions themself.

Through this project, I learned valuable interactive data visualisation skills. If I were to do this again, I might try to find a more complete dataset that doesn't have any null values. I also might try to incorporate more complex data analysis potentially with some data modelling.

9 Appendix

See 5 design sheets below

Sheet 1

1) Ideas

1. Choropleth map displaying total FV offences, with pie chart overlayed on each lga displaying breakdown of offence type
2. Two side-by-side choropleth maps, one displaying FV offences and the other displaying non-FV offences
3. Choropleth map displaying increase in offence rates between 2016 and 2025, user can select offence type
4. Choropleth map displaying offence rate. Interaction menu allows user to select one or more offence types, a year or range of years, FV or non-FV
5. Scatterplot showing offence rate over time for the whole state, each offence has a different coloured line, FV and non-FV distinguished by point shape
6. Bar chart with offence type on x-axis, offence rate on y-axis. Use colour to distinguish between FV and non-FV offences
7. Bar chart showing increase in offence rate between 2016 and 2025, with offence type on x-axis. Use colour to distinguish between FV and non-FV offences
8. Pie chart showing proportion of different offences out of total FV crime / total non-FV crime

2) Filter

Remove:

- Chart 8
  - Doesn't provide much depth of information
- Chart 2
  - Side by side choropleth maps make it difficult to compare values in the same LGA, especially in denser areas with smaller LGAs

3) Categorise

1. Location analysis, comparison of offence types
2. Comparison of FV vs non-FV offences, location analysis
3. Analysis of change over time, location analysis
4. Analysis of change over time, location analysis, comparison of offences types, comparison of FV vs non-FV offences
5. Analysis of change over time, comparison of offence types, comparison of FV vs non-FV offences
6. Comparison of FV vs non-FV offences, comparison of offence types
7. Analysis of change over time, comparison of offence types, comparison of FV vs non-FV offences

4) Combine and Refine

Group 1: Charts 1, 5

- Shows geographical distribution of total FV offence rate, along with breakdown of offence types and offence rates over time

Group 2: Charts 4, 7

- Shows geographical distribution of each offence, with interaction to allow user to filter data shown. Also shows increase in each offence rate between 2016 and 2025

Group 3: Charts 3, 6

- Shows geographical distribution of the increase in offence rates, along with a comparison of FV and non-FV offences

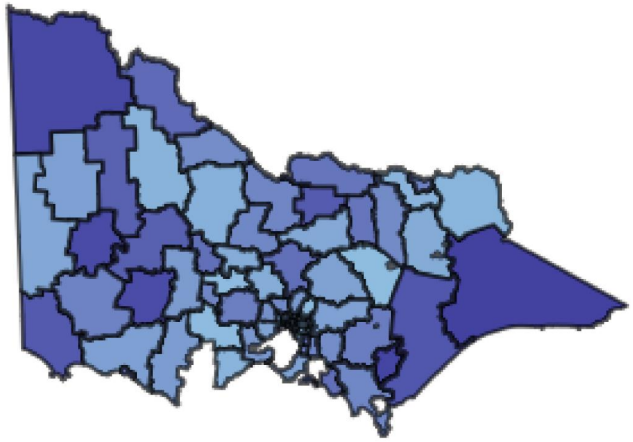
5) Question

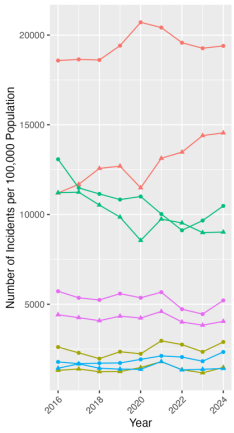
Do the three refined groups each provide enough information to answer the research questions?

- Group 3 doesn't show offence rates for each year, only the overall increase

Sheet 2

Layout





Number of FV Offences per 100,000 population

400 800 1200 1600

Select Year:

2016 2019 2022 2025

Type
- FV
- Non-FV

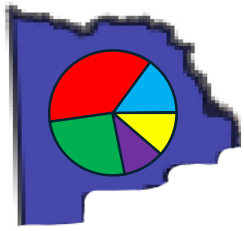
Incident
- Serious Assault
- Common Assault
- Stalking
- Harassment and Private Nuisance
- Threatening Behaviour

Operations

- Zoom in on map to view areas with smaller LGAs (ie, Melbourne)
- Click/hover over LGA to reveal pie chart showing proportions of different FV offences
- User can select which year to display on the map

Focus

Pie chart pops up when user clicks on LGA:



User will also be able to zoom in on different parts of the map

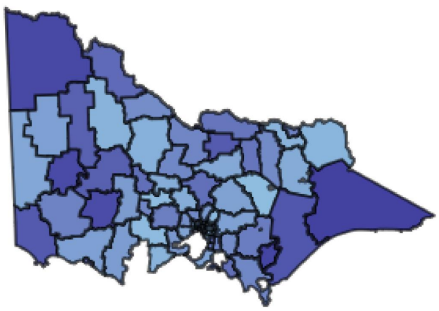
Discussion

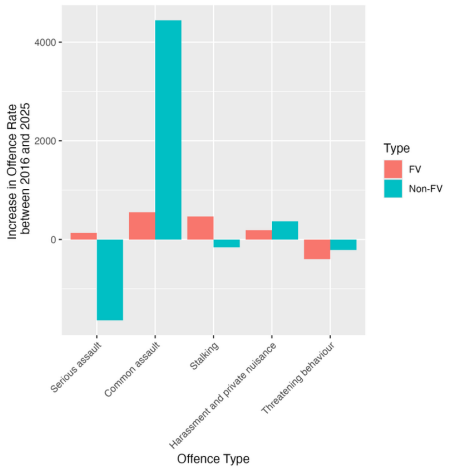
Advantages:
- User can compare total FV offence rates for different years
- Easy to compare rates of different offences for the whole state over time (scatterplot)

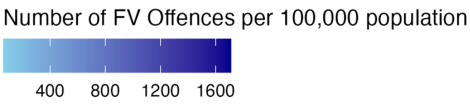
Disadvantages:
- No interaction with scatterplot
- No visual link between scatterplot and map
- Pie charts are difficult to compare, as the total rate is different for each LGA

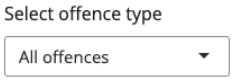
Sheet 3

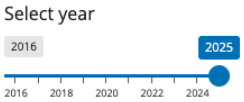
Layout











Discussion

Advantages:

- User can compare increase in FV and non-FV offence rates for the whole state
- Easy to compare offence rates for different LGAs, years, and offence types

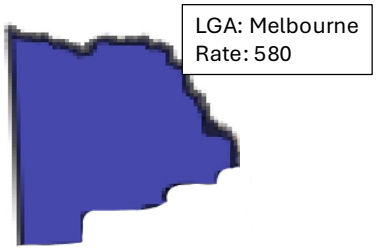
Disadvantages:

- No interaction with bar chart
- No visual link between bar chart and map
- Can't compare FV and non-FV offences for each year or LGA, only overall increase for whole state

Operations

- Zoom in on map to view areas with smaller LGAs (ie, Melbourne)
- Click/hover over LGA to reveal LGA name and offence rate
- User can select which year to display on the map
- User can select which offence to display on the map

Focus

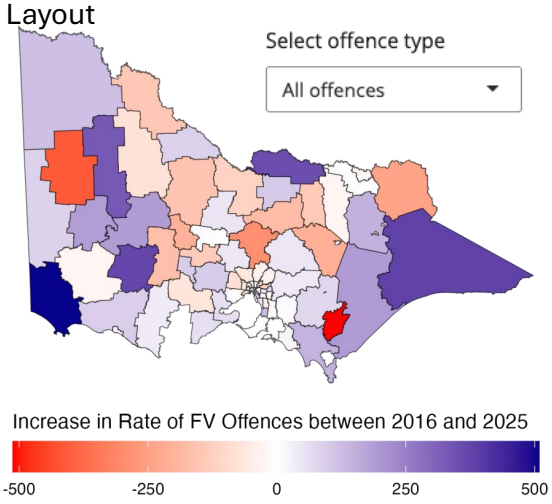


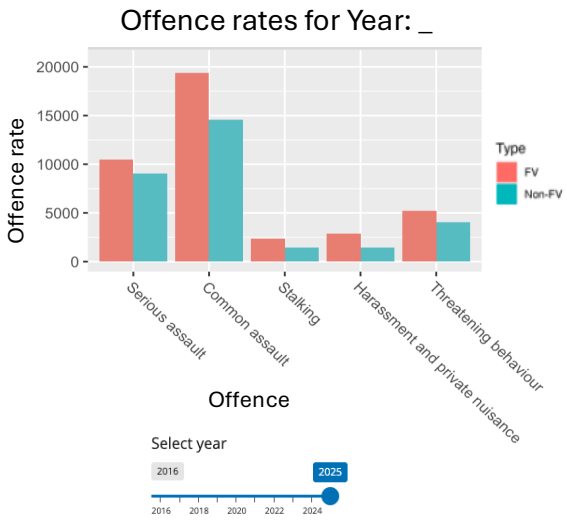
Box pops up when user clicks/hovers over an LGA

User will also be able to zoom in on different parts of the map

Sheet 4

Layout

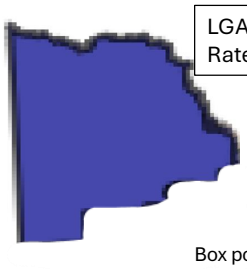




Operations

- Zoom in on map to view areas with smaller LGAs (ie, Melbourne)
- Click/hover over LGA to reveal LGA name and increase in offence rate
- User can select which offence to display on the map
- User can select which year to display on the bar chart

Focus



Box pops up when user clicks/hovers over an LGA

User will also be able to zoom in on different parts of the map

Discussion

Advantages:

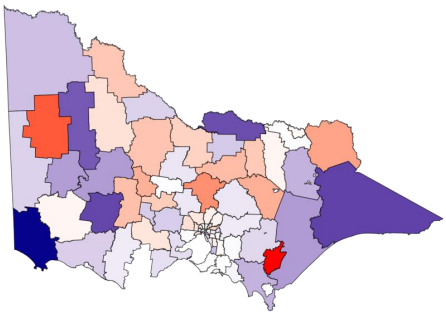
- User can compare FV and non-FV offences for a given year for the whole state
- User can compare different offences for a given year for the whole state
- Easy to compare increase in offence rates for different LGAs and offences

Disadvantages:

- No visual link between bar chart and map
- No common interaction between the bar chart and map
- Can't compare offence rates for different LGAs in different years, only the increase between 2016 and 2025

Sheet 5

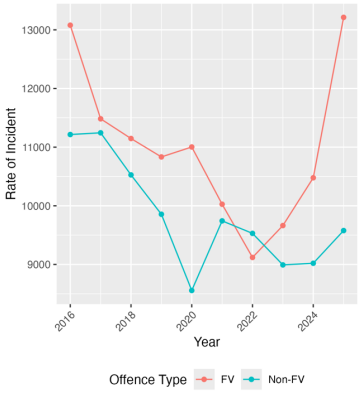
Layout



Increase in Rate of FV Offences between 2016 and 2025

-500 -250 0 250 500

LGA: (insert name here)



Click to reset scatterplot to show whole state data

Map Type:
- ○ Offence rate
- ✓ Increase in offence rate from 2016 to 2025

Year:

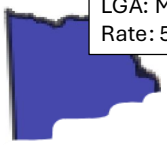
2016 2019 2022 2025

Offence:
- ✓ Serious assault
- ○ Common assault
- ○ Stalking
- ○ Harassment and private nuisance
- ○ Threatening behaviour
- ○ All offences

Operations

- Hover over LGA to show LGA name and offence rate / offence rate increase
- Click on LGA to show data for that LGA only on the scatterplot
- Button allows user to reset scatterplot to show data for the whole state
- Option menu allows user to show either offence rate or increase in offence rate on map
- Option menu allows user to select a year to display on the map (only applicable if user selects offence rate as map type)
- Option menu allows user to select an offence, or all offences to display on the map and scatterplot

Focus



LGA: Melbourne
Rate: 580

Box pops up when user clicks/hovers over an LGA

User will also be able to zoom in on different parts of the map



The above colour scheme will be used for the offence rate choropleth map (sky blue for low rate, dark blue for high rate)

Detail

Summary statistics to compute:
- Increase in offence rate between 2016 and 2025 for each LGA and offence
- Yearly offence rate for the whole state, both total FV offence rate and rate for each individual offence
- Yearly total offence rate for all FV offences, yearly total offence rate for all non-FV offences. Both for each LGA and the whole state

Can use RShiny and leaflet to create all elements of the visualisation, including interactions.

10 References

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